Rising Temperatures and Native Tree Resilience in New England

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Executive Summary

Background

The New England region, which includes Vermont, New Hampshire, Maine, Massachusetts, Connecticut, Rhode Island, and occasionally northern New York, is witnessing historic warming levels that have resulted in the region already surpassing benchmarks set by third party agencies. The effects of climate change also extend beyond temperature increases, with precipitation and seasonal patterns already beginning to fluctuate. The impact of this change is being felt by forest ecosystems and the environment as a whole.

Problem Summary

The New England region is rapidly warming due to climate change and is facing a variety of challenges because of it. Research shows that the region could see temperature increases of up to 7.6 °F, with widespread ramifications resulting from this change (Janowiak et al, 2018). The first way in which climate change will affect the region is tourism and impact to local populations. The skiing industry — which has long been an economic driver in many parts of New England — is no longer as stable as it used to be, with decreased snowfall and warmer winters shortening skiing seasons (Brown, 2025). Another way that climate change is impacting the region is through the direct impact of warming on ecosystems. Increased prevalence of droughts, extreme heat days, and high precipitation days create a more volatile environment that is less conducive to tree growth (Young and Young, 2021). Conversely, this extreme weather can encourage the presence of harmful pests that feed on and kill tree species, which further endangers native tree species (Orwig et al, 2012). Finally, new and existing tree diseases are becoming more prevalent, directly threatening the survival of many iconic trees that define the region (Shepherd et al, 2025).

Solution Summary

Two solutions are outlined, both of which impact key stakeholders and are able to be feasibly completed within a reasonable timeframe. The first solution [insert here]. The second solution is to assess a wide variety of tree species that can be used as new foundation species for adaptive forests. The results of research contained within this study highlight several key findings that must be taken into account when performing this kind of assessment. Primarily, studies that test the adaptivity of tree species must be conducted in real-world environments, as controlled studies introduce confounding variables that muddle results.

Background

Climate change is a pervasive issue across the United States and the world, but the rising temperatures has impacted New England particularly harshly. The New England region, which includes Vermont, New Hampshire, Maine, Massachusetts, Connecticut, Rhode Island, and occasionally northern New York, has already passed the 1.5 (degree)C warming level and has almost reached the 2 (degree) C level (Young and Young 2021, Young and Young 2025). The Intergovernmental Panel on Climate Change set the 1.5 degree level as “do not pass,” meaning the New England region is rapidly warming to dangerous levels (Masson-Delmotte et al, 2018). Alongside rising temperatures, New England is also predicted to experience significant changes in precipitation patterns, notably an increase in total precipitation and a decrease in number of wet days (Picard et al. 2023). This trend is also reflected in seasonal patterns, with spring starting sooner and seasonal differences increasing each year (Brown 2025).

These climatic changes are impacting New England’s forests. While responses vary from species to species, fruit and grain plants were significantly impacted by extreme temperatures during the reproductive stage (Hatfield and Prueger 2015).

Species like the European beech (*Fagus sylvatica*) and downy oak (*Quercus pubescens*) require winter temperatures to fall below a certain level in order to release their winter buds, and they will be impacted by the warming winters (Didion-Gency et al. 2024). Likewise, the decrease in soil moisture caused by the rising temperatures negates any positive impacts the rise in temperatures would otherwise have on tree species (Didion-Gency et al. 2022, McMillan 2023).

The eastern hemlock (*Tsuga canadensis*) is one of the foundation species for eastern North American forests, and its loss would disrupt ecosystems across New England (Ellison et al. 2005). The eastern hemlock is threatened by the presence of the hemlock woolly adelgid, which currently doesn’t pose a disastrous threat to eastern hemlocks, but the adelgids are predicted to thrive in the warmer winters New England is facing in the coming years, putting the eastern hemlocks in jeopardy (Orwig et al. 2012).

The American Beech (*Fagus grandifolia*) is another prevalent species in New England, and it is projected to increase in importance in coastal areas in coming years (Busby et al. 2009). The American Beech is currently suffering from the Beech leaf disease plaguing the eastern United States, which is killing many trees (Shepherd et al. 2025).

Our goal is to direct conservation efforts and guide further research into which species to prioritize and what strategies work best. Regan-Loomin (2025) has implemented a strategy of removing the direct competitors of trees that are more resilient to disease and climate change.

Problem Description

The New England region is rapidly warming due to climate change, with temperature increases outpacing those of other regions in the United States (Young and Young, 2025). While most warming has occurred since the 1980s, an acceleration in recent years has been observed, and the effect of this warming is wide reaching. Winter periods are facing the brunt of this change, with minimum and nighttime temperatures increasing and snow cover dramatically decreasing. Consequently, several key New England industries, such as the Skiing industry, for example, have been severely impacted (Brown, 2025). As temperatures increase in southern regions, snowfall — and skiing — has shifted northward, reducing income and affecting local communities that once relied upon that tourism traffic for survival.

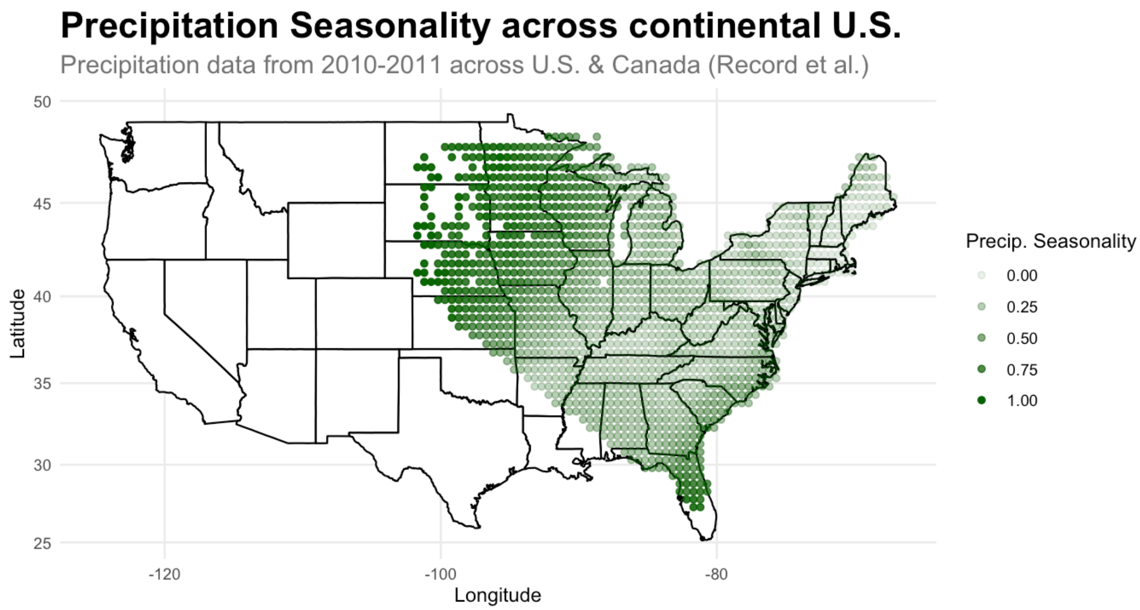
The degree to which the New England region will warm is still yet to be determined. However, previous research has identified a low-emission and high-emission scenario for the region, with projected temperature increases by the end of the century reaching 2.6 °F and 7.6 °F, respectively (Janowiak et al, 2018). Under the high emission scenario, extreme heat events (days with maximum temperature exceeding 90 °F) could increase by 16 to 60 days per year. Likewise, high rainfall events are projected to become a more common occurrence, underscoring the increase in weather extremes expected due to climate change.

A major effect of this changing climate is the unpredictability of weather in coming years. Young and Young (2021) described how the New England region's four-season climate is slowly being eroded. This not only has ramifications for the winter season, as outlined above, but also for the summer season. Drought is becoming increasingly prevalent, which has enabled tree-harming diseases and pests such as the spongy moth (Brown, 2025). Outbreaks of the spongy moth and other insects like the hemlock woolly adelgid have been more frequent in recent years, and research suggests that the prevalence of these attacks will only increase as climate change progresses (Orwig et al, 2012).

Another disease affecting tree populations in the region is beech leaf disease. Because of *Fagus Grandifolia*'s dominance in the northeast, the rise of beech leaf disease is particularly worrisome. A relatively new disease, beech leaf disease has been shown to dramatically decrease beech tree populations, especially in areas where beech trees are highly concentrated (Shepherd et al, 2025). Beech leaf disease has been shown to be most present when climate factors are stable (Selvi et al, 2026). This coincides with natural distributions of beech trees, as most beech populations lie within a low-precipitation seasonality band.

Figure 1

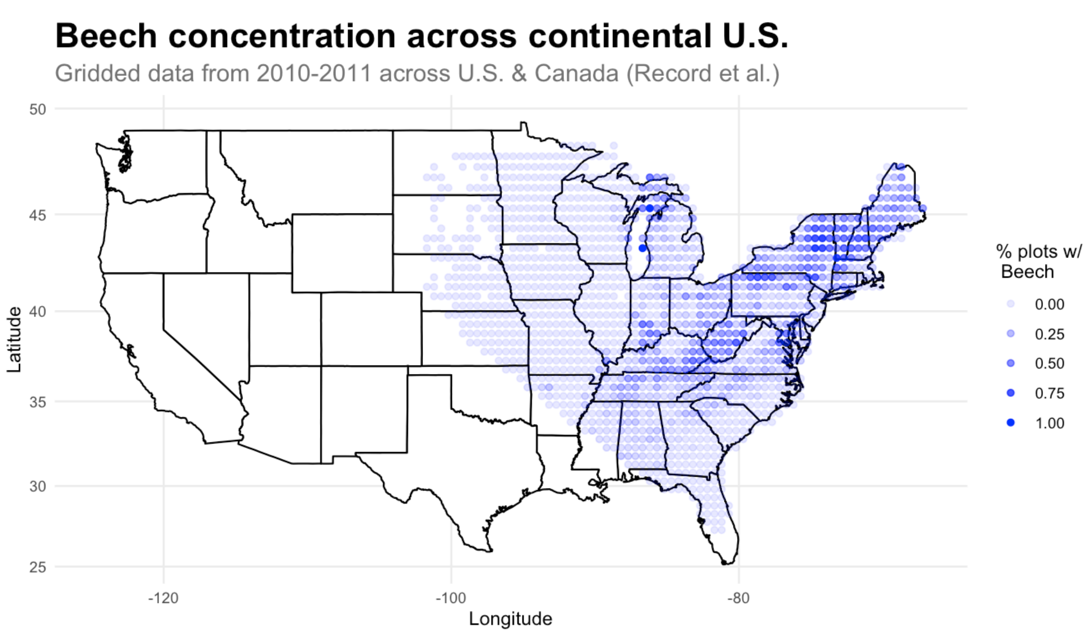
Precipitation seasonality across the continental U.S.



Note. The seasonality index describes relative degree of precipitation seasonality, with higher indices indicating a more volatile moisture climate.

Figure 2

Beech concentration across the continental U.S.



As seen in Figure 1, a band of consistent precipitation seasonality stretches from lower Appalachia to the Northeast. In Figure 2, beech tree concentration is shown, with a strong overlap of high beech presence existing along the stable band. As such, trees like the American beech could be at risk, as diseases like the beech leaf disease are thriving in the tree's natural habitat. Such situations could push the beech tree northward, with warming summers in southern regions exacerbating this trend.

Overall, several challenges face trees in the New England region. Firstly, warming climates are directly impacting tree populations and ecosystems. Second, warming climates are enabling insects that threaten tree species. Lastly, native tree diseases are continuing to spread, and the intersection between climate factors and already present diseases could spell disaster for old growth forests that have been around for centuries. This dramatic threat for native tree species is not only consequential for residents of the New England region, but also for forest conservationists, reforestation experts, and researchers. Conservationists interested in preventing the decrease of these old-growth forests will be facing major challenges. Likewise, reforestation experts will be interested in what trees are most suitable to replace those now-gone forests, and more research is needed in this area. Finally, researchers will look to identify the factors that are causing tree decline as well as why certain species are most suited to survive in a warming climate. Each of these factors requires adequate monitoring and extensive research, especially as the climate continues to rapidly change in this region.

Doing nothing — failing to perform tests, survey tree counts, identify new foundation species, and assess threats — will result in a society unequipped to deal with a foreign climate. As forests disappear and ecosystems form anew, a lack of understanding regarding what is occurring and how could be catastrophic. Conversely, by implementing the above methods, the New England region will be more prepared for weathering the coming storm, both literally and metaphorically.

Solution Description

The proposed solutions to the aforementioned problems should address one or more of the stakeholder concerns and should be able to be feasibly completed in the coming decades. The nature of climate-related goals is such that many targets are frequently abandoned or not reported on after the initial announcement. A study into corporate emissions goals found that nearly 40% failed to achieve the goal or stopped reporting on it (Jiang et al, 2025). While the below solutions will be, in majority, driven by public institutions, the need for accountability and feasibility of completion is integral to solution success. Additionally, without a direct, grounded link to the communities that the solution will be impacting, any such plan will be either doomed to fail or create adverse consequences. These factors are thus integral for evaluating potential solutions, and the proposed solutions below are ones that match the above criteria.

Solution #1: Implement widespread pre-commercial thinning practices that maintain diversity of forests while prioritizing the growth of healthy, resilient trees

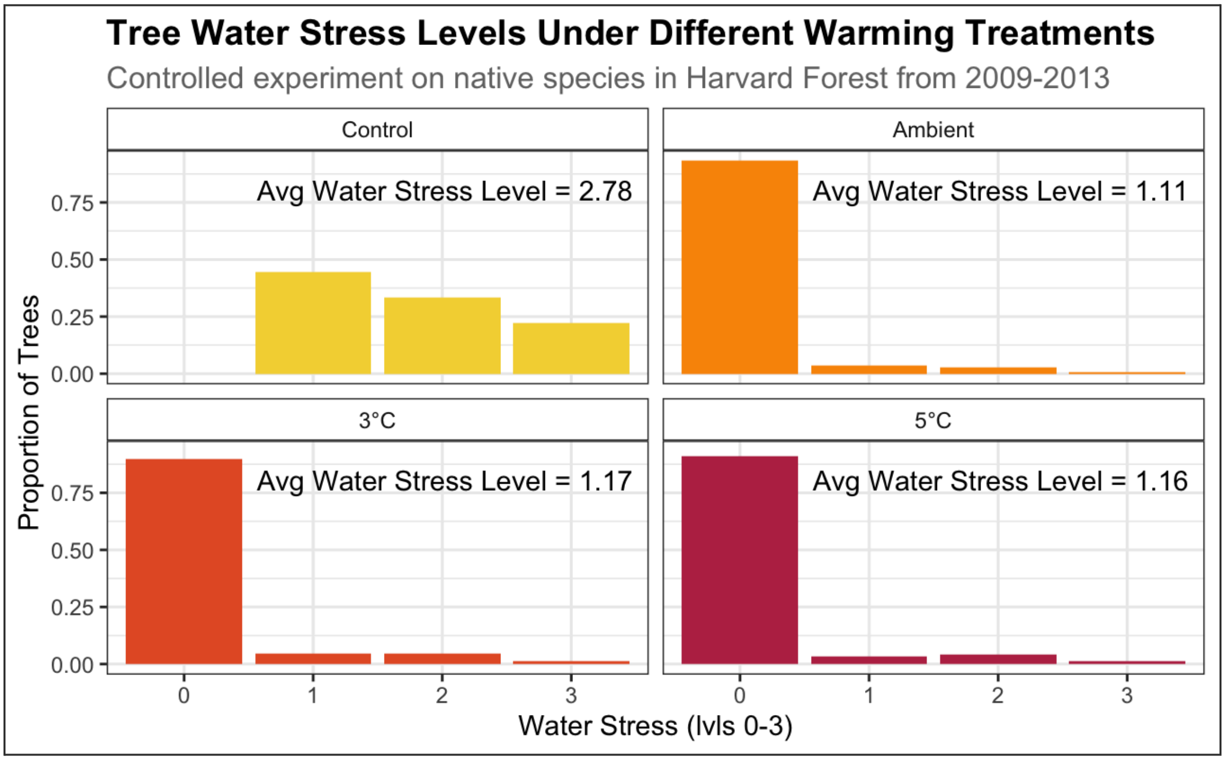
Pre-commercial thinning is a forest management method that has in the past been used by timber harvesters so they can plant the most trees possible and artificially select for the strongest, healthiest ones later, as opposed to depending on the growth and health of fewer trees (Emmingham et al, 1984). Recently, Great Mountain Forest (GMF) in Connecticut has begun using the practice to attempt to conserve their forest by prioritizing the growth of trees that have been most resilient to recent climate changes and disease outbreaks (Regan-Loomis, 2025). While they specify that they are attempting to preserve the diversity of the ecosystem and avoid a monoculture, the reality of the situation means that if disease outbreaks are concentrated on certain species, those species will be thinned at a larger scale than others. In the Great Mountain Forest, this is demonstrated most strongly with beech trees, which have been under attack by beech leaf disease for the last decade. According to Elizabeth Ward at the Connecticut Agricultural Experiment Station (CAES), "In Connecticut, every single beech tree has beech leaf disease," (Linden 2026). Most beech trees are thus being removed from GMF stands. This is the downside of pre-commercial thinning: while it is an adaptation to a changing climate and changing ecosystems, it does nothing to address the root causes of these shifts, preventing the death of the entire forest while allowing the structure of the ecosystem to be irreversibly altered.

Solution #2: Assess a wide sample of tree species to identify robust alternatives for the New England region that can withstand increasingly unpredictable climates

The loss of a foundation species in a forest ecosystem can be cataclysmic (Ellison et al, 2025). As trees like the eastern hemlock face increasing challenges, new species will need to be identified that are resistant to these threats. Several factors must be taken into account when identifying those species. Firstly, research into the effects of certain factors on tree growth must be completed in real-world environments. The result of our research shows that controlled in-chamber experiments failed to capture many of the climate effects present in the real world, and may have introduced artificial stability that benefitted tree growth even when conditions were considered "extreme". For example, the "water stress" variable present in Melillo et al. (2023) was set to zero in over 90% of cases for all of the in-chamber trees, while the control group — which was exposed to the outside environment — showed no zero scores and nearly a quarter of control trees were assigned the highest water stress score.

Figure 3

Tree water stress levels under different warming treatments



This discrepancy highlights a key difference between controlled experiments and real world ecology. Trees that are tested in closed environments are not being evaluated for their suitability to different climate scenarios, they are being evaluated for their suitability to different labs. Our analysis of the differences between Record and Ellison (2023) and Melillo et al. (2023) thus suggests that tree species should be thoroughly evaluated through real world methods that avoid artificially introduced confounding variables.

This solution would benefit key stakeholders by supporting researchers and reforestation experts. The results of such an assessment could be used to plan for coming deforestation as a result of climate change and could also inform local efforts to continue ongoing tree planting initiatives.

Recommendations

Recommendation?: **Perform ecological experiments to identify climate factor causality on tree growth in real-world environments**

Conclusions

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above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty Edited by Science Officer Science Assistant Graphics Officer.
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